Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/04/2024
PI Name:	Dove, Adrienne Ph.D.		
Project Title:	Processing and Analysis of the Strata-1 Dataset to Explore Granular Mechanics of a Realistic System in a Long-Duration Low-Gravity Environment		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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PI Email:	adrienne.dove@ucf.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	407-823-3136
Organization Name:	Department of Physics		
PI Address 1:	University of Central Florida		
PI Address 2:	4000 Central Florida Blvd, PS 430		
PI Web Page:			
City:	Orlando	**State:**	FL
Zip Code:	32816-2385	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
Contact Email:	louise.s@nasa.gov		
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Task Description:	The exploration of small Solar System bodies such as asteroids and moons is essential to understanding the origins of our Solar System and extending our reach beyond the Earth. To date there have been few flight experiments that have explored the behavior of layered regolith materials in microgravity, especially complex material assemblages and long-term microgravity environments. The Strata series of experiments is advancing the state of the art by observing the evolution of granular materials in microgravity and vibrational environments. Through data from a long-duration flight on the International Space Station (ISS) we can explore the behavior of granular materials in order to provide context for, and interpretation of, observations of the surfaces of small bodies, as well as provide a rich dataset that can be used to validate numerical simulations. This in turn will also provide a new avenue for acquisition of mission-enabling data for both scientific and exploration missions to a wide range of Solar System bodies. Our objectives for the proposed research project are to: 1. Assess the granular dynamics of particle systems with known size distributions as a function of their acceleration environment. To do this, we will (a.) characterize the long-term evolution of regolith sorting in an environment (microgravity and vibrational) that is relevant to that on the surfaces of asteroids; (b.) understand the behavior of different particle systems (idealized particles, realistic simulants) in these asteroid-like environments; and (c.) characterize the structures that form and evolve in columns of regolith, and how those are influenced by acceleration forcing in microgravity. 2. Develop an empirical relationship between gravitational and acceleration profiles and size sorting behavior (velocity, particle size distribution) in regolith. As part of this, we will determine how this relationship fits within established models of regolith size sorting mechanisms. To meet these objectives, we will: 1) Process and analyze particle motion and distributions in the Strata imagery, available in the Physical Sciences Informatics (PSI), for the: a. Tubes with glass shards of known particle size distribution b. Tubes with spherical glass beads c. Tubes with carbonaceous meteorite simulant and meteoritic material 2) Process, sync, and analyze the coincident Space Acceleration Measurement System (SAMS) accelerometer data to provide context for the behavior seen in the imaging analysis. Relevance The proposed investigation is relevant to the PSI because it seeks to analyze data from the Strata-1 ISS experiment. This investigation falls under the research area of Materials Science, and we aim to use the collected data to investigate the processes at work on materials and conditions representative of the surfaces of asteroids and other small, airless bodies in the solar system, and as well as other surfaces dominated by granular materials. Through analysis of flight-based experiments performed in microgravity, we will explore the validity of granular segregation hypotheses that may be used to explain asteroid surface particle (boulder vs. regolith) size distributions. We will develop empirical models that can be used to inform numerical simulations of granular segregation on asteroids, as well as to interpret remote observations of asteroid surfaces and their evolution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research investigation to analyze data from the Strata-1ISS experiment, which was built and flown to explore granular material behavior in reduced gravity. We aim to use the collected data to investigate the processes at work on materials and conditions representative of the surfaces of asteroids and other small, airless bodies in the solar system, and as well as other surfaces dominated by granular materials. Understanding these behaviors is relevant to other physical and materials science questions about granular material behavior in response to different gravity and forcing functions. Through analysis of flight-based experiments performed in microgravity, we will explore the validity of granular segregation hypotheses that may be used to explain asteroid surface particle (boulder vs. regolith) size distributions. We will develop empirical models that can be used to inform numerical simulations of granular segregation on asteroids, as well as to interpret remote observations of asteroid surfaces and their evolution. This may help with future technologies intended to interact with planetary surfaces and functions in reduced gravity.
	Image processing We initially developed a procedure to process the "shard" tube data, which was the most straightforward due to the red-green-blue (RGB) color selection of the particles and their behavior in the tubes. In the last year, we have done more refinement of that code, and have applied similar analysis to the spherical particle tube, which also has approximately RGB particles. As previously mentioned, a significant limitation of the spherical particle tube analysis is that the smallest (red) spheres completely dominate the images, likely due to these smallest particles electrostatically clinging to the walls. This thus reduces visibility for some of the internal particle behavior for a significant portion of time, although some trends do emerge. We can now both observe and break down the imagery of the 2D view of the changes in the particle distributions inside of the tubes for the full lifetime of the experiment. Note that we have now successfully tied the video data to real time (each camera metadata has its own random date) so that the behaviors observed in the tubes can be aligned with the accelerometer data. See more details below. We are continuing analysis to normalize the data, look at variance over time, and make more direct comparisons between the sphere and shard tubes. We do see significant differences in the behavior of the small shards and the small spheres, with the shards sticking less to the tube than the spheres; this likely has to do with the amount of surface area/contact area between the particles and the sides of the tubes, and we are working on some calculations to estimate the magnitudes of the forces that are necessary to liberate both types of particles. Some observations that we are currently exploring more are the trends in particle spacing and motion with respect to location in the tube, especially now that we are able to compare behaviors in the two tubes. We are also working on identifying alternative ways to classify the behavior of the materials in the tubes. Accelerometer data Data from the International Space Station (ISS) Space Acceleration Measurements System-II (SAMS-II) accelerometer that was mounted on the front of the Strata-1 experiment is downloaded from the publicly available website. We have been working to develop a way to process the massive amounts of accelerometer data available over the year of the Strata-1 mission to identify features that may be of relevance to the tube data, and especially to look for correlations in “events” at certain dates and times. This may include looking at long-term trends; but, for now, the focus has been on identifying higher magnitude acceleration events as a function of frequency and time. We are now working on

Task Progress:	cross-correlating those with the trends seen from the shard and sphere tube analyses. We utilize Fourier transform techniques to filter and de-noise the data, as well as identify features as functions of frequency for the x-, y-, and z-axis channels. We then search for dates and times when the magnitude of the acceleration is above a certain threshold. These "events" are identified; we determine a magnitude and duration of the event (i.e., "is it still high at that frequency for an extended period or is it transient"?). We then find the peak frequencies contributing to this high-magnitude signal. The undergraduate student that was working on this graduated in May 2023, and a new student has been coming online more recently but has not yet made significant additional progress. More recently, we have corresponded with Ken Hrovat (Zin Technologies) about the Microgravity Acceleration Measurements System (MAMS) data, which gives more information about the quasi-steady microgravity regime and will be used to help understand the gravity gradient and rotational effects on the experiment. Additionally, we have data on the types of large-scale events that occur on the ISS, such as reboosts, thrusters, and inversion events. Similar to the SAMS data, we are working on aligning these data to the image data, identifying changes due to events and attempting to understand the general forcing function of the gravity data on the tubes and particles. This work is being developed into a manuscript. Dissemination Initial results from this project have been included in the following presentations: • A. Dove, A. Nicola, K. Schang, D. Cassidy, 2023, "Granular mechanics behaviors observed in multiple variable-gravity experiments relevant to planetary surfaces", AGU Annual Meeting, San Francisco, CA. Abstract EP23B-03 • A. Dove, K. Schang, and M. Scott (2023) Lunar and Asteroid-Relevant Environments Across Suborbital Flight Platforms. Next-Generation Suborbital Researchers Conference, Broomfield, CO • K. Schang et al., 2022, "Strata-1 Image Analysis: Understanding Granular Dynamics of Regolith Materials", AGU Fall Meeting, Chicago, IL • M. Scott et al., 2022, "Processing and Analysis of ISS Accelerometer Data", AGU Fall Meeting, Chicago, IL • A. Dove et al., 2022, "Lunar and Asteroid-Relevant Environments Across Suborbital Flight Platforms", NSRC, Broomfield, CO (focused on related experiments, including Strata-1 and analysis) • This work has also been presented at the UCF Research Symposium (April, 2023), where undergraduate student Michael Scott presented on his work with the ISS SAMS acceleration measurements. [Note: See Bibliography for complete citations. (Ed., 4/5/24)]
Bibliography Type:	Description: (Last Updated: 04/05/2024)
Abstracts for Journals and Proceedings	Dove A, Nicola A, Schang K, Cassidy D. "Granular mechanics behaviors observed in multiple variable-gravity experiments relevant to planetary surfaces." AGU 23 (Annual Meeting of the American Geophysical Union), San Francisco, CA, December 11-15, 2023. Abstracts. AGU 23, Annual Meeting of the American Geophysical Union, San Francisco, CA, December 11-15, 2023. Abstract: EP23B-03. https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1446963 , Dec-2023
Abstracts for Journals and Proceedings	Dove A, Schang K, Scott M. "Lunar and asteroid-relevant environments across suborbital flight platforms. " 2023 Next-Generation Suborbital Researchers Conference, Broomfield, CO, February 27 - March 3, 2023. Abstracts. 2023 Next-Generation Suborbital Researchers Conference, Broomfield, CO, February 27 - March 3, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Schang K et al. "Strata-1 image analysis: Understanding granular dynamics of regolith materials." AGU Fall Meeting, Chicago, IL, December 12-16, 2022. Abstracts. AGU Fall Meeting, Chicago, IL, December 12-16, 2022. , Dec-2022
Abstracts for Journals and Proceedings	M. Scott et al. "Processing and analysis of ISS accelerometer data." AGU Fall Meeting, Chicago, IL, December 12-16, 2022. Abstracts. AGU Fall Meeting, Chicago, IL, December 12-16, 2022. , Jan-2022
Abstracts for Journals and Proceedings	Dove A, Schang K, Scott M. " Lunar and asteroid-relevant environments across suborbital flight platforms." 2022 Next-Generation Suborbital Researchers Conference, Broomfield, CO, February 27 - March 1, 2022. Abstracts. 2022 Next-Generation Suborbital Researchers Conference, Broomfield, CO, February 27 - March 1, 2022. , Feb-2022